

Zero-WAM: In-Context World-Action Modeling from Human Videos for Open-Ended Task Generalization

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<https://robbyant-research.github.io/Zero-WAM/>

Abstract

Zero-shot cross-task generalization, where a policy must execute manipulation tasks never seen during training, remains a central challenge in robot learning. In large language models, a novel task can be performed simply by specifying it in the context, without any parameter update. This form of in-context learning (ICL) turns generalization into a problem of task specification. To achieve cross-task generalization, we bring this paradigm to robotic manipulation, and argue that the natural task specification for manipulation is a human video: unlike language, it provides rich visual cues about the intended task evolution. We present **Zero-WAM**, a causal video-action model that executes unseen tasks by following in-context human video guidance. To address the scarcity of task-rich paired human-robot data, we propose an automatic pipeline that converts task-sampled robot trajectories into semantically matched human videos, yielding *HumanGen*, a dataset of 74.2K human-robot ICL pairs across 8.6K tasks. For model training, we further introduce an in-context future chunk prediction (IFP) objective that suppresses shortcuts learned from seen tasks and forces the policy to draw task information from the video prompt. On seven unseen tasks in RoboTwin 2.0 simulation, Zero-WAM achieves a 47.0% average success rate, an absolute improvement of 29.5 percentage points over the strongest video-action baseline. In real-world evaluations, it follows human video guidance to generalize to unseen task configurations involving multi-object scenes, long-horizon manipulation, and fine-grained insertion.

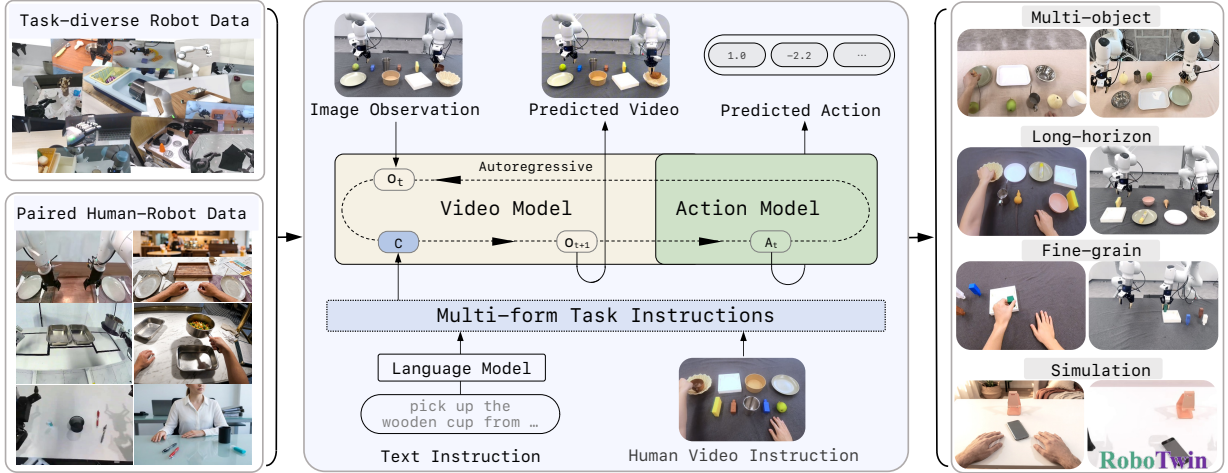


Figure 1 Overview of Zero-WAM. Zero-WAM predicts future robot videos and executable actions from an in-context human video prompt or a language instruction. The human video instructions are generated at scale from task-sampled robot trajectories, alongside a task-balanced robotic video-action pre-training corpus. We validate the effectiveness of Zero-WAM on unseen simulated tasks and complex real-world tasks.

1 Introduction

Zero-shot cross-task generalization is essential for general-purpose robotic manipulation: a robot should infer how to act in a task it has not practiced, using only information available at deployment. Inspired by in-context learning (ICL) [1, 2], where a model solves a new problem from input context without parameter updates, we view robotic task generalization as specifying an unseen task through deployment-time context. This perspective suggests a route toward open-ended task generalization: the policy could infer the intended task from context and translate it into executable robot behavior.

Recent vision-language-action (VLA) models [3, 4, 5, 6, 7, 8, 9, 10] and video-action models [11, 12, 13, 14, 15, 16] have substantially expanded the capabilities of robot policies, yet both predominantly rely on language as the task interface. Language, however, often underspecifies manipulation tasks: spatial constraints, intermediate states, and temporal structure can be cumbersome to articulate, and even detailed instructions provide no direct visual evidence of how the scene should evolve. Human demonstration videos provide a natural in-context specification of the intended task [17, 18]. By directly presenting the desired visual state changes and their temporal evolution, they provide concrete visual evidence of how the scene should evolve. Although a human video does not provide executable robot actions, it gives the policy a visual reference from which to infer the desired task evolution and realize it through the robot’s own embodiment and dynamics.

Exploiting human videos at scale, however, presents two key challenges. First, large-scale, task-rich human-robot paired data remain scarce and expensive to collect manually [19, 20, 21]. Learning from human video instructions requires semantically corresponding robot trajectories that retain executable actions, yet such pairings are rarely available at scale. Second, a policy trained on seen tasks can learn shortcuts that allow it to ignore the in-context video. In particular, the next robot video-action chunk can often be predicted from the robot history and text instruction alone. Consequently, the model may perform well on familiar training tasks without learning to use the human video, and then underuse the video at test time, precisely when it is needed to specify an unseen task.

To address these challenges, we introduce **Zero-WAM**, a causal video-action model that follows in-context human video instructions for zero-shot robotic task generalization. Zero-WAM supports two forms of task specification within a single policy: language instructions and human demonstration videos. Given either form of instruction, the model autoregressively predicts future robot videos and executable actions. Human videos enable Zero-WAM to condition these predictions on demonstrated visual state changes, providing information about the intended task evolution beyond that available from language and robot history alone.

To scale training with human video instructions, we propose an **in-context human video generation pipeline** that uses task-sampled robot trajectories to construct semantically matched human manipulation videos. Each human video is paired with its corresponding robot trajectory, which retains the executable robot actions, to form a human-robot in-context learning (ICL) pair. The resulting *HumanGen* dataset contains 74.2K human-robot ICL pairs spanning 8.6K tasks. We further perform task-balanced curation of robotic pretraining data by repartitioning public robot trajectories into more than 6,000 manipulation tasks and sampling trajectories at the task level. This procedure yields approximately 400K robot trajectories per training epoch, which we refer to as *Task-diverse VA*. Task-balanced sampling both supplies task-rich source trajectories for the in-context human video generation pipeline and prevents video-action pretraining from being dominated by repeated teleoperation trajectories from a small number of tasks. Together, *HumanGen* and *Task-diverse VA* provide scalable human video task specifications and diverse robot video-action dynamics for causal video-action pretraining. Figure 2 summarizes the resulting data composition and the in-context human video generation pipeline.

To ensure that Zero-WAM uses the in-context human video rather than relying on shortcuts from robot history and text, we introduce an in-context future chunk prediction (IFP) objective. Standard next-chunk prediction can often be solved using only local robot history, particularly for tasks observed during training. IFP instead supervises multiple strided chunks of future robot video from the current robot-video representation, encouraging the model to encode the longer-term task evolution conveyed by the human video.

We evaluate Zero-WAM on zero-shot cross-task generalization in RoboTwin 2.0 [22] simulation and on real-world robotic manipulation. In RoboTwin 2.0, Zero-WAM achieves a 46.95% average success rate across seven unseen tasks, outperforming LingBot-VA [11] by an absolute margin of 29.50 percentage points.

Real-world experiments further demonstrate human-video-guided generalization to unseen task configurations involving multi-object scenes, long-horizon manipulation, and precision-demand insertion, with Zero-WAM outperforming LingBot-VA across all three task families.

Our contributions are summarized as follows:

- We formulate zero-shot robotic task generalization as in-context world action modeling, where a single causal policy supports both language and human videos as task instructions.
- We propose a scalable in-context human video generation pipeline that automatically converts task-sampled robot trajectories into semantically matched human video instructions, yielding the *HumanGen* dataset of 74.2K human-robot ICL pairs over 8.6K tasks; the same task-level sampling also yields *Task-diverse* VA data, a task-balanced corpus for autoregressive robotic video-action pre-training.
- We propose Zero-WAM with an in-context future chunk prediction (IFP) objective that discourages shortcut learning from seen-task trajectories and strengthens the use of in-context human video prompts.
- We demonstrate effective zero-shot cross-task generalization in RoboTwin, where Zero-WAM substantially improves over video-action baselines, and further evaluate real-world generalization to unseen task configurations without collecting corresponding robot data or updating any model parameters.

2 Data Curation

Zero-shot robotic task generalization requires an instruction interface that transfers to unseen tasks, together with training data that covers diverse task dynamics rather than merely increasing the number of robot trajectories. As an engineering foundation, we first curate *Task-diverse* VA data by re-sampling public robotic pre-training data at the task level (Section 2.1). Building on the same task-level sampling, we introduce the **in-context human video generation pipeline** to convert task-sampled robot videos into human video instructions (Section 2.2). The generated human-robot ICL pairs form *HumanGen*, which includes Pre-train ICL (External), Pre-train ICL (In-house), Simulation ICL, and Real-world ICL (Section 2.3). Figure 2 summarizes this data composition and the pipeline used to generate *HumanGen* data.

2.1 Task-Diverse Video-Action Data

Task-diverse VA data are curated from public robotic video-action (VA) pre-training datasets, including AgiBot [23], InternData-A1 [24], Open-X-Embodiment [4], RoboCOIN [25], and RoboMIND [26]. These source datasets are the same public VA pre-training datasets used by LingBot-VA [11]. Although these datasets contain large numbers of robot trajectories and broad task coverage, many trajectories come from repeated tele-operation of the same task. Using the raw distribution directly would make pre-training overly influenced by redundant trajectories, preventing the model from fully benefiting from the task diversity already present in the source datasets. We address this by re-partitioning each dataset into tasks, where each task is defined by the combination of the manipulation action and the object. The task labels are obtained from the original dataset metadata when available, or parsed from the robot trajectories when the metadata are insufficient. We then sample a bounded number of trajectories from each task, with the sampling bound adjusted according to the intra-task diversity of each source dataset. Across the five source datasets, we sample more than 6,000 tasks and approximately 400K corresponding robot trajectories in each training epoch. This balanced video-action corpus is effective for adapting a general-domain video generation model into an autoregressive robotic video-action model.

2.2 In-Context Human Video Generation Pipeline

Human video demonstrations convey task semantics directly through visual state changes, are easier to obtain than robot demonstrations, and do not depend on the robot embodiment used at test time. Scaling task-rich human-robot in-context learning pairs is therefore an effective route to cross-task generalization.

Manually collecting such paired human and robot action data is difficult, and the cost becomes especially prohibitive when high task diversity is required. We therefore start from the robotic pre-training corpus and



Figure 2 Data construction and in-context human video generation. Top: *Task-diverse VA* data provide task-balanced robotic video-action pre-training data, while *HumanGen* contains Pre-train ICL (External), Pre-train ICL (In-house), Simulation ICL, and Real-world ICL pairs. Bottom: the in-context human video generation pipeline converts task-sampled robot videos into human video instructions.

again sample trajectories at the task level, following Section 2.1. This gives us a broad set of robot videos with executable action annotations. We then introduce an **in-context human video generation pipeline** that converts these robot trajectories into human manipulation videos with the same task semantics. To increase the diversity of visual alignment in the human-robot ICL pairs, the generated human videos include variations in background, viewpoint, environment style, object instance, and object placement while preserving the same task semantics.

The bottom part of Figure 2 shows the in-context human video generation pipeline for generating human video instructions. For each sampled robot video, we first use a VLM (Gemini 3.1 Pro [27] or Qwen3.6-Plus [28]) for task analysis. The VLM extracts task-level information, including the task name, initial object states, object state changes, and final object states. It also produces an image-editing prompt that transforms the first robot video frame into the initial observation of a human manipulation scene. We inject the visual-alignment variations described above through this image-editing prompt while preserving the task semantics. Given the first robot frame and the image-editing prompt, an image editing model (Nano Banana 2 [29] or Qwen-Image-2.0 [30]) generates the initial human observation image for the corresponding task. We then use the VLM to process this edited human observation together with the extracted object-state information, producing a video-generation prompt that describes how human hands should manipulate the objects to complete the

Table 1 Comparison with existing task-level paired human-robot datasets. “Multi-source” indicates whether the dataset is constructed from multiple data sources. “Embod.” is the number of robot embodiments. “Human view” specifies the viewpoint of human videos. “Human acquisition” indicates whether the human videos require manual collection or are automatically generated. “Align. div.” indicates whether human-robot pairs include diverse degrees of visual alignment, such as changes in background, viewpoint, environment style, object instance, or object placement. “Samples” is the number of human video instructions, and “Tasks” is the number of task categories.

Dataset	Multi-source	Embod.	Human view	Human acquisition	Align. div.	Samples	Tasks
MIME [19]	✗	1	third	manual	✗	8.3K	20
EgoMimic [20]	✗	1	ego	manual	✗	2.1K	3
BC-Z [17]	✗	1	ego	manual	✓	18.7K	100
EgoHumanoid [33]	✗	1	ego	manual	✗	1.2K	4
RH20T [21]	✗	4	ego & third	manual	✗	110K	147
EgoScale [34]	✗	1	ego	manual	✗	10.3K	344
HumanGen (ours)	✓	>45	ego & third	auto-generated	✓	74.2K	8.6K

task. This prompt is sent to a video generation model (Wan 2.7 [31] or Kling AI 3.0 [32]) to synthesize the human manipulation video. Finally, the VLM evaluates each generated video for task semantic preservation and physical plausibility, and qualified human videos are paired with their original robot trajectories as ICL samples.

2.3 In-Context HumanGen Dataset

HumanGen is a collection of human-robot ICL pairs generated with the in-context human video generation pipeline. Each pair contains a generated human video instruction and its corresponding robot trajectory with executable actions. *HumanGen* is organized by data source into Pre-train ICL (External), Pre-train ICL (In-house), Simulation ICL, and Real-world ICL. Following the same task-diverse sampling principle used to curate *Task-diverse VA* data, source robot videos are sampled by task rather than by raw trajectory frequency, so that the ICL data are not dominated by repeated executions of a small set of tasks.

Pre-train ICL (External). This subset is built from the same five public robotic video-action datasets used for *Task-diverse VA* data: AgiBot [23], InternData-A1 [24], Open-X-Embodiment [4], RoboCOIN [25], and RoboMIND [26]. These datasets cover diverse robot embodiments, scenes, and objects, including more than 45 robot embodiments. We sample robot trajectories by task from each dataset and convert them into semantically matched human videos using the in-context human video generation pipeline. Specifically, we sample 3,354 tasks and 6,660 trajectories from AgiBot, 261 tasks and 12,515 trajectories from InternData-A1, 438 tasks and 9,290 trajectories from Open-X-Embodiment, 512 tasks and 6,513 trajectories from RoboCOIN, and 497 tasks and 6,210 trajectories from RoboMIND. In total, Pre-train ICL (External) contains 5,062 tasks and 41,188 human-robot ICL pairs. For this subset, we intentionally diversify the visual alignment between the human and robot videos: we enforce changes in the scene environment, tabletop background, object instances, and object placement, and balance first-person and third-person viewpoints. Because task semantics are preserved across these visual variations, the model is pushed to learn human-robot task correspondences that are invariant to changes in scenes, objects, and viewpoints, improving its robustness to visually diverse ICL prompts.

Pre-train ICL (In-house). To further increase task coverage, we sample robot trajectories by task from our in-house robotic datasets. This subset covers multiple robot embodiments, such as bimanual Franka and Galaxea R1 Pro, and contains 3,522 tasks and 30,247 human-robot ICL pairs. When using the in-context human video generation pipeline to produce the corresponding human videos for this subset, we reduce the ratio of third-person viewpoints and decrease the frequency of scene, object, and object-placement changes. This produces more visually aligned ICL samples while maintaining task diversity.

Simulation ICL. To support cross-task evaluation in simulation, we generate ICL data for 50 RoboTwin tasks [22]. This subset contains 2,500 ICL samples, with 50 samples for each task. Among them, 43 tasks are used for post-training (2,150 training samples), while the remaining 7 tasks are reserved as unseen tasks for

zero-shot cross-task evaluation.

Real-world ICL. Real-world ICL contains human-robot pairs collected on the bimanual Franka evaluation embodiment. It includes 252 human-robot ICL pairs across the three real-world task families used in our evaluation: 120 pairs from 30 object-to-container placement training task combinations, 96 pairs from 16 three-object sequential manipulation training task combinations, and 36 pairs for two-table-leg insertion. This subset is used for post-training so that the model can fit the evaluation robot kinematics for cross-task evaluation.

Comparison with existing datasets. Table 1 compares *HumanGen* with existing task-level paired human-robot datasets. The statistics are taken from their official papers. Unlike prior datasets that rely on manual human data collection, *HumanGen* uses the in-context human video generation pipeline to automatically generate semantically matched human videos from robot trajectories, enabling scalable construction of human-robot ICL pairs. *HumanGen* also combines multiple data sources (public, in-house, simulation, and real-world), covers substantially more robot embodiments, contains human data from both ego and third-person views, and contains paired data with diverse degrees of visual alignment. Most importantly, *HumanGen* contains substantially broader task coverage, which is a key factor for zero-shot cross-task generalization.

3 Zero-WAM: In-Context World Action Modeling

In zero-shot cross-task manipulation, the robot must execute a novel task without finetuning. We pursue this ability with a world action model (WAM) pretrained on large-scale human-robot ICL pairs and task-balanced robotic video-action data: at deployment, Zero-WAM uses either a language instruction or a human video demonstration as the task specification to drive execution of unseen tasks.

3.1 Preliminaries

Flow matching for video generation. For video generation, flow matching [35] learns a velocity field along a continuous path between a clean video and Gaussian noise. Given a clean video $\mathbf{x}_0$, noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and flow time $t \in [0, 1]$, the noised video and target velocity are

$$\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\epsilon, \quad \mathbf{v}_t^* = \epsilon - \mathbf{x}_0. \quad (1)$$

With condition c , the flow-matching loss is

$$\mathcal{L}_{\text{fm}} = \mathbb{E}_{\mathbf{x}_0, \epsilon, t} \left[\|\mathbf{v}_\theta(\mathbf{x}_t, t, c) - \mathbf{v}_t^*\|_2^2 \right]. \quad (2)$$

At inference, a video is generated by starting from noise and integrating the learned velocity field back to $t = 0$.

Causal video-action modeling. We follow the causal video-action framework of the LingBot-VA series [11, 12], in which each control step is a causal prediction problem: given the task condition and history observation, the model predicts the next robot video chunk together with its temporally aligned executable action chunk.

Each robot trajectory is chunked into $\tau = \{(\mathbf{x}^i, \mathbf{a}^i)\}_{i=1}^N$, where $\mathbf{x}^i$ is a video chunk and $\mathbf{a}^i$ is the temporally aligned action chunk. We use chunk-level notation throughout the method. In implementation, each video or action chunk is encoded into modality-specific representations before entering the Transformer, and the attention mask operates over these tokenized representations. We use c to denote the task condition, which can be a language instruction or a same-task human video instruction. At chunk index i , the causal model predicts the next video chunk $\mathbf{x}^{i+1}$ and the aligned action chunk $\mathbf{a}^{i+1}$:

$$p_\theta(\mathbf{x}^{i+1}, \mathbf{a}^{i+1} \mid \mathbf{x}^{\leq i}, \mathbf{a}^{\leq i}, c). \quad (3)$$

This joint prediction is factorized into video prediction followed by action decoding:

$$\begin{aligned} p_\theta(\mathbf{x}^{i+1}, \mathbf{a}^{i+1} \mid \mathbf{x}^{\leq i}, \mathbf{a}^{\leq i}, c) &= p_\theta^{\text{vid}}(\mathbf{x}^{i+1} \mid \mathbf{x}^{\leq i}, \mathbf{a}^{\leq i}, c) \\ &\quad \cdot p_\theta^{\text{act}}(\mathbf{a}^{i+1} \mid \mathbf{x}^{\leq i}, \mathbf{a}^{\leq i}, \mathbf{x}^{i+1}, c), \end{aligned} \quad (4)$$

where p_{θ}^{vid} predicts the next video chunk, and p_{θ}^{act} serves as an inverse dynamics model that decodes executable actions from the predicted next robot video chunk. During training, the model is given the ground-truth trajectory prefix $(\mathbf{x}^{\leq i}, \mathbf{a}^{\leq i})$ and is supervised on the next video-action chunk. For trajectories from *Task-diverse* VA data, the condition is the language instruction, $c = \ell$. For trajectories from *HumanGen* data, the condition is $c = \{\mathbf{h}, \ell\}$, where $\mathbf{h}$ is the same-task human video instruction and ℓ is the language instruction.

Mixture-of-Transformers Design. We implement p_{θ}^{vid} and p_{θ}^{act} as a video Transformer and an action Transformer under a Mixture-of-Transformers (MoT) design. Each modality has its own parameters, including separate QKV projections, FFNs, and output heads, while the video and action representations remain in a single sequence and interact only through the shared attention layers. Within this sequence, the action chunk representations are placed after the future video representations, so that action decoding can attend to the predicted future robot video, as required by the factorization above.

Model instantiation. Following LingBot-VA, we instantiate Zero-WAM by converting *Wan-2.2-TI2V-5B* [36], a bidirectional text/image-to-video generation model, into the causal video-action policy described above. Built on this framework, the remainder of this section presents the components specific to Zero-WAM: human video as in-context task specification (Section 3.2) and in-context future chunk prediction (Section 3.3).

3.2 Human Video as Task Specification

For an unseen task, fully specifying the desired behavior in language is often difficult, and the instruction must further be grounded into task dynamics that the policy has never observed. We therefore adopt human video as an in-context task specification that presents the desired dynamics directly: at deployment, a human video demonstrating the unseen task serves as the instruction. To teach the policy to follow this interface, we train on the *HumanGen* dataset, which is constructed by sampling source robot trajectories at the task level (Section 2). Each semantically matched pair contains a generated human video $\mathbf{h}$, which serves as the visual task specification, and its corresponding robot trajectory, which provides supervised future video and action chunks. Since the human video and paired robot trajectory may differ in embodiment, viewpoint, background, and object placement, the model must learn task-level correspondence instead of copying motions from human videos.

During training, we apply a teacher-forcing attention mask over the tokenized representations of human video chunks, robot video chunks, and action chunks. The human video $\mathbf{h}$ is prepended before the robot trajectory and serves as prefix memory for robot video prediction. At chunk index i , the video Transformer predicts the next robot video chunk $\mathbf{x}^{i+1}$ from the context $\mathcal{C}^{\text{vid}, i}$:

$$\mathcal{C}^{\text{vid}, i} = [[\mathbf{h}, \mathbf{x}^{\leq i}], \mathbf{a}^{\leq i}, \ell], \quad (5)$$

$$p_{\theta}^{\text{vid}}(\mathbf{x}^{i+1} | \mathcal{C}^{\text{vid}, i}) = p_{\theta}^{\text{vid}}(\mathbf{x}^{i+1} | [\mathbf{h}, \mathbf{x}^{\leq i}], \mathbf{a}^{\leq i}, \ell). \quad (6)$$

For the next robot action chunk prediction, the action Transformer predicts $\mathbf{a}^{i+1}$ from robot-domain history and the predicted next robot video chunk, following the inverse-dynamics factorization in Section 3.1. It does not directly attend to the human video $\mathbf{h}$: the task semantics conveyed by $\mathbf{h}$ are already absorbed into the predicted next robot video chunk, so action decoding reduces to standard inverse dynamics. The action-prediction condition therefore remains unchanged from standard robot video-action training:

$$\mathcal{C}^{\text{act}, i} = [\mathbf{x}^{\leq i}, \mathbf{a}^{\leq i}, \mathbf{x}^{i+1}, \ell], \quad (7)$$

$$p_{\theta}^{\text{act}}(\mathbf{a}^{i+1} | \mathcal{C}^{\text{act}, i}) = p_{\theta}^{\text{act}}(\mathbf{a}^{i+1} | \mathbf{x}^{\leq i}, \mathbf{a}^{\leq i}, \mathbf{x}^{i+1}, \ell). \quad (8)$$

During training, $\mathbf{x}^{i+1}$ in the action context is the ground-truth next robot video chunk under teacher forcing; at inference, it is replaced by the video chunk generated by the video Transformer.

RoPE [37] offset for ICL human video. ICL human video and multi-view robot videos are encoded by the same VAE encoder, and therefore share the same visual latent space. Within the token sequence, we distinguish the ICL human video from robot videos through the height-axis RoPE coordinates: robot video latents keep their original coordinates, while human video latents are shifted by an offset along the height axis. Let (q, y, x) denote the temporal, vertical, and horizontal coordinate of a visual latent, and let H_{mv} be the height of the

multi-view robot video latent layout. We assign RoPE coordinates as

$$\text{pos}_{\text{robot}}(q, y, x) = (q, y, x), \quad (9)$$

$$\text{pos}_{\text{human}}(q, y, x) = (q, y + \Delta_H, x), \quad \Delta_H > H_{\text{mv}}. \quad (10)$$

The offset places the human video latents outside the coordinate range of the robot video latents, preventing representation confusion between the ICL human video and robot videos when they interact in the same sequence.

3.3 In-Context Future Chunk Prediction

For in-context samples, the model should infer the task evolution from the human video $\mathbf{h}$ and transfer it to the robot domain. However, during teacher-forcing training on seen tasks, the immediate next robot video chunk can often be predicted by extrapolating the recent robot history ($\mathbf{x}^{\leq i}$). This creates a shortcut that allows the model to reduce the training loss without learning to use the in-context human video. When the model is evaluated on unseen tasks, this shortcut causes it to continue relying on the robot history and underuse the ICL signal precisely when that signal is needed to specify the new task. To counter this shortcut, we draw inspiration from the multi-chunk prediction of Next Forcing [14] and introduce in-context future chunk prediction (IFP): a training-only auxiliary objective that requires the model to predict multiple strided future robot video chunks from the current robot-video representation.

Let $s \geq 1$ be the temporal stride and K be the number of future chunks to predict. The k -th future target is the robot video chunk $\mathbf{x}^{j_k}$, where the target index j_k is defined as

$$j_k = (i + 1) + 1 + (k - 1)s, \quad k = 1, \dots, K. \quad (11)$$

To predict these targets, we add K IFP modules $\{G_k\}_{k=1}^K$, where G_k is responsible for denoising the k -th strided future video chunk under the flow-matching objective. Each IFP module has the same architecture as a single Transformer layer in the video branch and is initialized from the last video Transformer layer.

IFP operates on the current robot-video representation produced by the main branch. When predicting the current robot video chunk $\mathbf{x}^{i+1}$ under $\mathcal{C}^{\text{vid}, i}$, we collect robot-video hidden representations from M intermediate layers of the main video Transformer. Let $\{\mathbf{r}_m^{i+1}\}_{m=1}^M$ denote the collected feature representations of $\mathbf{x}_t^{i+1}$, the noised chunk of $\mathbf{x}^{i+1}$ at flow time t . We concatenate these multi-layer representations and project them back to the single-layer hidden dimension with a lightweight MLP projection P_{fuse} :

$$\phi^{i+1} = P_{\text{fuse}}(\text{Concat}(\{\mathbf{r}_m^{i+1}\}_{m=1}^M)). \quad (12)$$

Each IFP module G_k predicts its strided future chunk $\mathbf{x}^{j_k}$ conditioned on this fused current representation, the clean robot video/action history, and the language instruction:

$$p_{\theta, k}^{\text{ifp}}(\mathbf{x}^{j_k} \mid \phi^{i+1}, \mathbf{x}^{\leq i}, \mathbf{a}^{\leq i}, \ell), \quad k = 1, \dots, K. \quad (13)$$

The IFP loss applies the flow-matching objective defined above to each strided target, with w_k denoting the loss weight for the k -th future video chunk target:

$$\mathcal{L}_{\text{ifp}} = \sum_{k=1}^K w_k \mathcal{L}_{\text{fm}}(\mathbf{x}^{j_k}; \phi^{i+1}, \mathbf{x}^{\leq i}, \mathbf{a}^{\leq i}, \ell). \quad (14)$$

Importantly, the IFP modules are not directly conditioned on the human video prompt $\mathbf{h}$. Their only access to its task information is through ϕ^{i+1} , which is extracted after the main robot-video Transformer has interacted with the in-context human video. If the IFP modules were directly conditioned on $\mathbf{h}$, the auxiliary branch could learn an independent human-video-conditioned future predictor without forcing the main robot-video Transformer to encode the in-context task information; since the IFP modules are removed at inference, the deployed policy would then gain nothing from the auxiliary supervision. We therefore condition IFP only on ϕ^{i+1} , so the future loss supervises the current robot-video representation produced by the main branch. Multiple future chunks are predicted in parallel with temporal stride, increasing the prediction difficulty without introducing temporal dependence among future chunk predictions during training.

3.4 Training and Inference

Training data mixture. We train Zero-WAM on a mixture of *Task-diverse VA* and *HumanGen* data: *Task-diverse VA* supplies diverse robot-domain video-action dynamics, and *HumanGen* provides human video task specifications grounded in executable robot actions. The two data families are drawn with sampling weights rather than in proportion to raw dataset sizes.

Training objective. For each target next video chunk $\mathbf{x}^{i+1}$, we sample noise ϵ^{i+1} and flow time t , and form

$$\mathbf{x}_t^{i+1} = (1-t)\mathbf{x}^{i+1} + t\epsilon^{i+1}, \quad \mathbf{v}_t^{*,i+1} = \epsilon^{i+1} - \mathbf{x}^{i+1}. \quad (15)$$

Given task condition c , the next-chunk video loss is

$$\mathcal{L}_{\text{fm}}^{i+1}(c) = \|\mathbf{v}_{\theta}^{\text{vid}}(\mathbf{x}_t^{i+1}, t \mid \mathbf{x}^{\leq i}, \mathbf{a}^{\leq i}, c) - \mathbf{v}_t^{*,i+1}\|_2^2. \quad (16)$$

For action chunk prediction, we use an analogous flow-matching objective in action space. Given Gaussian noise $\epsilon_a^{i+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and flow time $r \in [0, 1]$, we form

$$\mathbf{a}_r^{i+1} = (1-r)\mathbf{a}^{i+1} + r\epsilon_a^{i+1}, \quad \mathbf{u}_r^{*,i+1} = \epsilon_a^{i+1} - \mathbf{a}^{i+1}, \quad (17)$$

and optimize

$$\mathcal{L}_a^{i+1}(c) = \|\mathbf{u}_{\theta}^{\text{act}}(\mathbf{a}_r^{i+1}, r \mid \mathbf{x}^{\leq i}, \mathbf{a}^{\leq i}, \mathbf{x}^{i+1}, c) - \mathbf{u}_r^{*,i+1}\|_2^2. \quad (18)$$

For *Task-diverse VA* samples, the condition is language-only, $c = \ell$, and the loss is

$$\mathcal{L}_{\text{VA}} = \mathbb{E}_{i,t,r,\epsilon,\epsilon_a} [\mathcal{L}_{\text{fm}}^{i+1}(\ell) + \lambda_a \mathcal{L}_a^{i+1}(\ell)]. \quad (19)$$

For *HumanGen* samples, the video prediction is conditioned on $c = \{\mathbf{h}, \ell\}$ and augmented with in-context future chunk prediction (IFP). The action loss remains conditioned on ℓ , because the action Transformer does not directly attend to the human video:

$$\mathcal{L}_{\text{ICL}} = \mathbb{E}_{i,t,r,\epsilon,\epsilon_a} [\mathcal{L}_{\text{fm}}^{i+1}(c) + \lambda_a \mathcal{L}_a^{i+1}(\ell) + \lambda_{\text{ifp}} \mathcal{L}_{\text{ifp}}]. \quad (20)$$

Inference. At test time, the IFP modules are removed, and Zero-WAM supports two modes of task specification. In language-only mode, the model conditions on the text instruction, i.e., $c = \ell$. In ICL mode, the human video is encoded once and its tokens are cached as prefix memory, so the model conditions on $c = \{\mathbf{h}, \ell\}$, where the language instruction ℓ is optional. In both modes, the model first generates the next robot video chunk and then decodes the aligned action chunk.

4 Experiments

4.1 Implementation

Implementation details. Following LingBot-VA [11], we instantiate **Zero-WAM** from *Wan-2.2-TI2V-5B* [36] and convert the image-to-video generation backbone into the causal video-action model described in Section 3. The video Transformer follows the Wan-2.2 backbone with hidden dimension $d_v = 3072$ and 30 transformer layers. The action Transformer uses hidden dimension $d_a = 3072$, with parameters initialized from the video branch, and is connected with the video Transformer through the MoT architecture. Both streams use RoPE positional encoding, and the height-axis RoPE offset for ICL human video is set to $\Delta_H = 32$. We use the Wan-2.2 VAE to encode robot videos and human videos into latent representations. The language instruction is encoded with T5 [38]. The IFP module predicts $K = 4$ future robot video chunks with temporal stride $s = 2$. The corresponding future-prediction loss weights are $(w_1, \dots, w_4) = (0.5, 0.25, 0.15, 0.15)$.

Training details. We train Zero-WAM with the flow-matching video loss, action loss, and IFP loss defined in Section 3. We use the AdamW optimizer [39] with peak learning rate 1×10^{-4} and weight decay 0.01. During pre-training, *Task-diverse VA* data and *HumanGen* data are drawn with a sampling ratio of 1:5. For non-ICL samples, we drop the language instruction with probability 0.1. For ICL samples, we drop the ICL



Figure 3 Unseen tasks in RoboTwin 2.0 simulation. These tasks involve unseen-object manipulation, articulated-object manipulation, bimanual manipulation, and long-horizon manipulation.

human-video latent with probability 0.1, and increase the language-instruction dropout from the Wan-2.2 default of 0.1 to 0.4, reducing dependence on language when the model learns from human video instructions. During pre-training, the robot video chunk size is randomly sampled from 1 to 4. The pre-training takes 15,360 GPU hours. Following LingBot-VA, each GPU packs samples with different token lengths into one sequence and uses attention masks to isolate different samples, which improves training throughput while keeping the maximum token length per GPU within 160K.

Inference details. The pre-trained Zero-WAM supports two inference modes. In language-only mode, the model conditions on the language instruction without the ICL human video, and the video CFG scale is set to 5. In ICL mode, the model conditions on the ICL human video and disables the language instruction. We use ICL classifier-free guidance with guidance scale 5 [40]. For both modes, the inference chunk size is fixed to 2, and the action CFG scale is set to 1.0.

4.2 Simulation Experiments

Setting. We evaluate zero-shot cross-task generalization in the clean setting of RoboTwin 2.0 simulation [22]. RoboTwin 2.0 contains 50 bimanual manipulation tasks with 50 robot trajectories per task. For every trajectory, we generate a corresponding human video instruction with the proposed **in-context human video generation pipeline**, forming the Simulation ICL subset of *HumanGen* data. To evaluate cross-task generalization, we split RoboTwin 2.0 at the task level: 43 tasks are used for post-training and 7 tasks are reserved for evaluation as unseen tasks, so the evaluated tasks do not appear as robot demonstrations during training. The unseen tasks¹ are *place object on scale*, *stamp seal*, *open microwave*, *move stapler to pad*, *place bread in basket*, *place empty cup*, and *stack blocks three*. As shown in Figure 3, these tasks cover pick-and-place tasks with unseen objects and target containers, unseen articulated-object manipulation, and unseen long-horizon manipulation.

Training. On RoboTwin 2.0, we initialize Zero-WAM from the checkpoint pre-trained on *Task-diverse VA* data and in-context *HumanGen* data. We train with 64 GPUs for 4,000 post-training steps. During post-training, we jointly sample from *Task-diverse VA*, *HumanGen*, and RoboTwin data at a ratio of 2:10:3. As in pre-training, each GPU packs multiple samples up to 160K tokens.

Baselines. We compare against two video-action baselines under the same cross-task protocol. Since Zero-WAM is initialized from Wan-2.2, we build a direct Wan-based baseline by initializing from *Wan-2.2-TI2V-5B*, adopting the same MoT causal video-action framework, and training only on the 43 seen RoboTwin tasks. We refer to this baseline as WAN-Action. We also compare against LingBot-VA, a leading video-action model, using its released pre-trained checkpoint and post-training it on the same seen tasks. Both baselines use only seen-task robot video-action trajectories during post-training, following the standard cross-task setup [41].

Evaluation. All experiments are evaluated over three random seeds. For each seed, we run 100 closed-loop rollouts per unseen task in simulation and compute the task success rate. Final results report the mean and standard deviation across the three seeds. For human-video-conditioned variants, the task is specified by a generated human video instruction; for language-only variants, the task is specified only by text.

¹For *stamp seal* and *move stapler to pad*, we slightly adjust the success criteria to make them more appropriate for cross-task generalization evaluation. We apply the same modified criteria to all methods to ensure a fair comparison and will release our modified task versions.

Table 2 Zero-shot cross-task results on seven unseen RoboTwin tasks. We report task success rates and their macro average over three evaluation seeds.

Task	WAN-Action	LingBot-VA	Zero-WAM
Place object on scale	3.00±2.16	6.17±4.87	24.67±2.05
Stamp seal	7.33±1.25	3.67±2.49	47.00±4.55
Open microwave	2.26±1.60	29.33±10.66	59.00±2.83
Move stapler to pad	10.67±1.70	23.33±8.22	69.14±2.93
Place bread in basket	15.26±2.55	17.33±6.18	35.00±3.74
Place empty cup	38.33±2.05	42.33±7.85	84.87±0.18
Stack blocks three	0.00±0.00	0.00±0.00	9.00±2.16
Average	10.98±1.07	17.45±1.40	46.95±0.72

Table 3 Real-world unseen-configuration evaluation. We report task success rates over 30 real-robot trials. LingBot-VA is evaluated with language instructions, while Zero-WAM uses human video instructions.

Task	Train combos	Train demos	LingBot-VA	Zero-WAM
Object-to-container placement	30	120	43.3	53.3
Three-object sequential manipulation	16	96	10.0	33.3
Two-table-leg insertion	–	36	0.0	16.7

Results. Table 2 summarizes the main RoboTwin results with one row for each unseen task. The three methods form an ordered comparison of what is learned before RoboTwin post-training: WAN-Action adapts the generic Wan-2.2 video prior using only the seen RoboTwin tasks; LingBot-VA additionally benefits from existing robotic video-action pre-training; Zero-WAM is further pre-trained on task-balanced robotic data and human-robot ICL pairs. On the seven unseen tasks, Zero-WAM achieves an average success rate of 46.95%, compared with 17.45% for LingBot-VA and 10.98% for WAN-Action, corresponding to improvements of 29.50 and 35.97 percentage points.

The improvement is consistent across the unseen task set rather than being driven by a single easy task. Zero-WAM outperforms both baselines on all seven tasks, with large gains on *open microwave* and *stamp seal*, where the policy must infer unseen articulated-object or relocation dynamics from the task specification. It also reaches 84.87% on *place empty cup*, more than doubling both baselines. *Stack blocks three* remains the hardest unseen long-horizon task, but Zero-WAM is the only method among the main comparisons that achieves non-zero success. These results indicate that human video instructions and task-balanced robotic pre-training jointly improve the model’s ability to execute unseen task dynamics.

4.3 Real-World Experiments

We validate the effectiveness of Zero-WAM on a real bimanual Franka robot. We evaluate three task families: *object-to-container placement*, *three-object sequential manipulation*, and *two-table-leg insertion*. For each task family, we collect a small set of seen-task robot demonstrations to adapt the policy to the real-robot kinematics, while holding out the evaluated task configurations. At test time, Zero-WAM conditions on the human video instruction alone, without a language instruction, and executes the unseen task configurations on the real robot, while the LingBot-VA baseline is provided with detailed textual task descriptions.

Object-to-container placement. The robot must pick an object and place it into a target container using either single-arm manipulation or bimanual coordination. We collect 120 seen-task demonstrations from 30 object-container training combinations to adapt the policy to the Franka embodiment. We hold out part of the object and container set for evaluation, so each test configuration contains at least one object or container not seen in the robot training demonstrations. Zero-WAM achieves 53.3% success, compared with 43.3% for language-conditioned LingBot-VA.

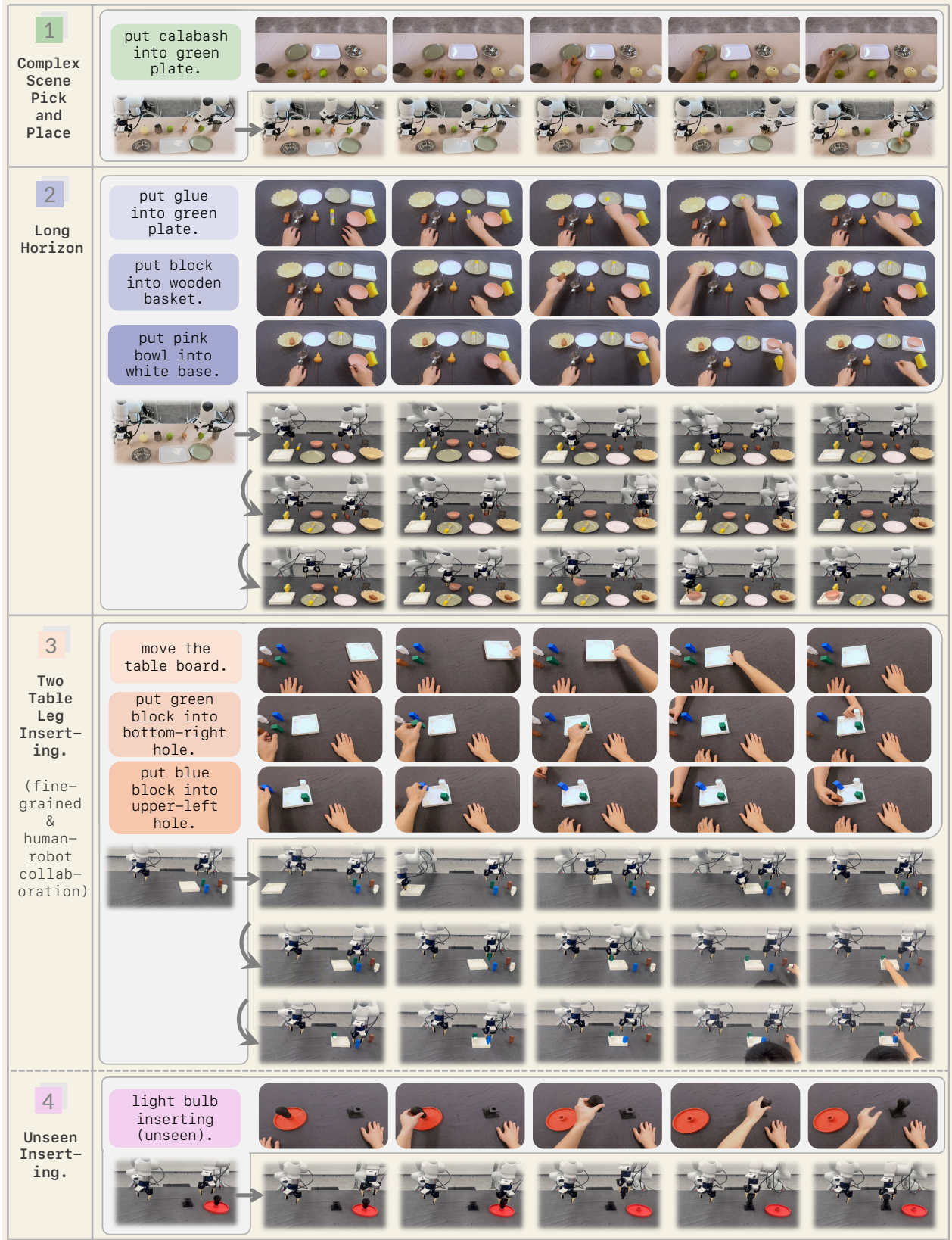


Figure 4 Qualitative real-world evaluations of Zero-WAM with human video instructions.

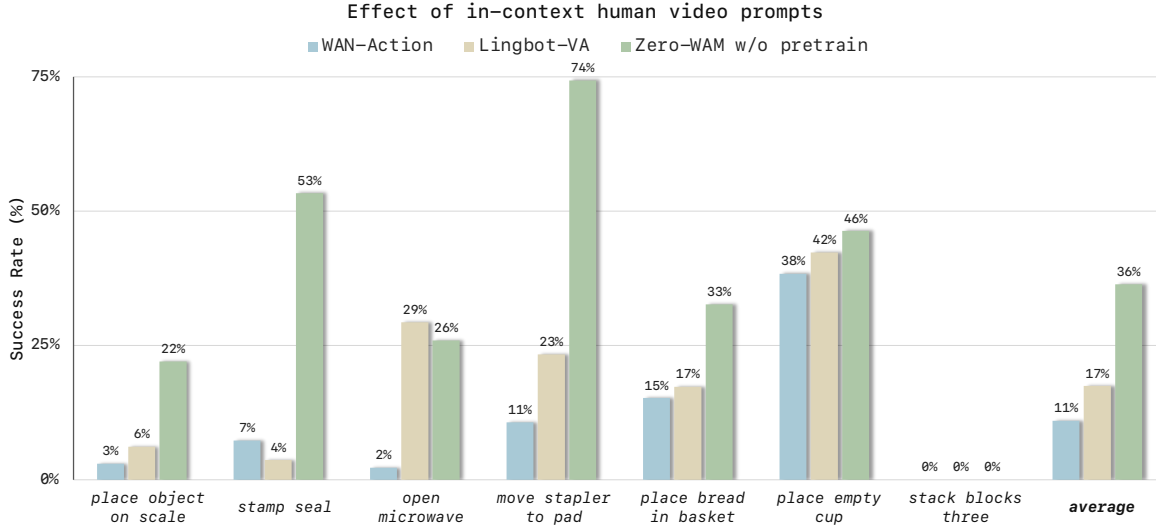


Figure 5 Effect of in-context human video prompts on RoboTwin unseen tasks. Bars report task success rates and their macro average.

Three-object sequential manipulation. This task evaluates whether the policy can follow a human video that specifies a long-horizon manipulation order. The scene contains multiple objects and containers, and each test episode selects three objects in a random order shown by the human video instruction. We collect 96 demonstrations from 16 training task combinations, for which we also collect corresponding human video instructions. At test time, the human video manipulates three objects in an arbitrary order, including unseen objects and containers; Zero-WAM reaches 33.3% success, substantially outperforming the 10.0% success rate of LingBot-VA.

Two-table-leg insertion. This task tests fine-grained human-video task specification. The task requires inserting two table legs into selected holes on a tabletop base, and the human video specifies which colored leg should be inserted into which target hole. We collect only 36 demonstrations for this precise insertion setting. At test time, the human video specifies the target holes for the two table legs, and the robot must complete both insertions accordingly. Although the absolute success rate is limited by the difficulty of precise physical insertion, Zero-WAM succeeds on unseen insertion configurations while LingBot-VA obtains zero success. We also observe successful qualitative transfer to table legs and tabletop bases with unseen colors and types, further indicating that fine-grained task specifications are followed more reliably from human video instructions than from language alone.

4.4 Ablation Studies

All ablations are conducted in RoboTwin with the same 43 seen tasks and 7 unseen tasks used in the main simulation experiment. RoboTwin enables large-scale closed-loop evaluation under a fixed task-level split, making the ablation comparisons more reliable than small real-world trial sets. The ablations isolate three factors in Zero-WAM: human video ICL, IFP, and task-balanced robotic pre-training.

Effect of in-context human video prompts. Figure 5 isolates the effect of ICL from large-scale pre-training by training all compared variants exclusively on data from the 43 seen RoboTwin tasks. Zero-WAM w/o pretrain is initialized from the Wan-2.2 pre-trained weights and trained with ICL human videos as instructions. WAN-Action is initialized from the same Wan-2.2 pre-trained weights but uses only the language instruction during RoboTwin training. LingBot-VA also uses the language instruction on RoboTwin, but retains its robotic video-action pre-training. Compared with WAN-Action, adding human video instructions raises the average success rate from 10.98% to 36.36%, showing that ICL provides task information beyond text-only conditioning. Zero-WAM w/o pretrain also substantially outperforms LingBot-VA, which achieves an average success rate of 17.45% despite its robotic video-action pre-training, further demonstrating the importance of human video instructions for specifying unseen tasks. However, all three variants obtain zero success on

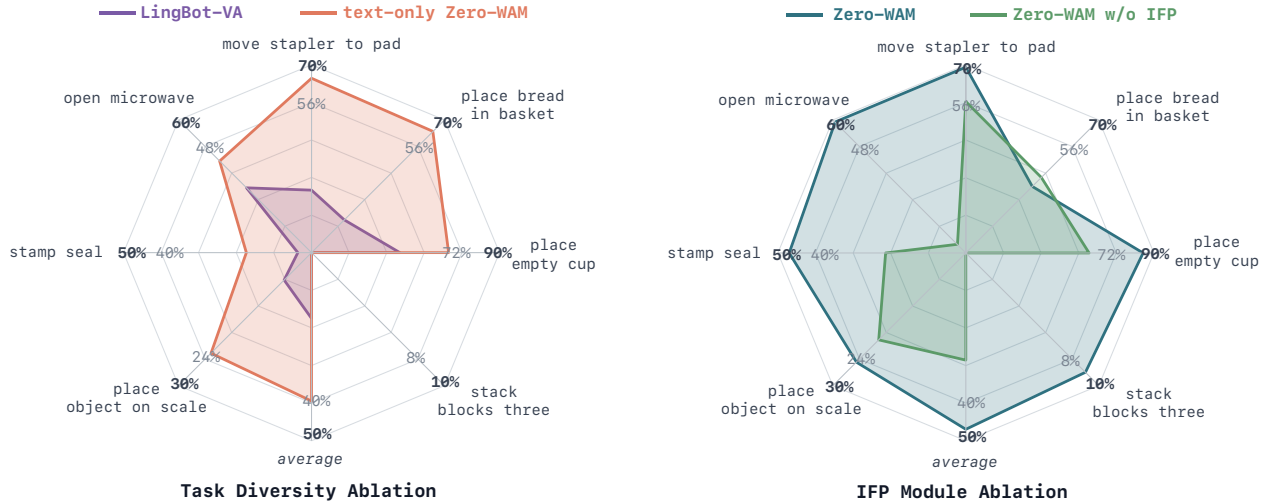


Figure 6 Ablations of task-balanced robotic pre-training and in-context future chunk prediction on RoboTwin unseen tasks. Left: task-balanced robotic data ablation, comparing LingBot-VA with a text-only Zero-WAM variant. Right: IFP module ablation, comparing full Zero-WAM with Zero-WAM w/o IFP. Radar axes indicate per-task success rates and the macro average.

stack blocks three, whereas the full Zero-WAM achieves non-zero success in the main experiment. This result suggests that small-scale ICL training data are insufficient for effectively solving unseen long-horizon tasks, highlighting the necessity of large-scale task-diverse ICL pre-training.

Effect of in-context future chunk prediction. When training with human videos as instructions, the model may still rely on the language instruction and past robot visual observations to infer the next robot video chunk, without sufficiently using the in-context human video prompt. Our IFP objective addresses this problem by supervising strided future robot video chunks from the current robot-video representation, thereby encouraging the main video branch to encode the task evolution conveyed by the human video. Figure 6 (right) shows that adding IFP improves the seven-task average from 28.55% to 46.95%. The gains are especially large on *open microwave*, an unseen articulated-object manipulation task, and *stamp seal*, an unseen rare-object relocation task. Notably, on *stack blocks three*, an unseen long-horizon manipulation task, IFP breaks the zero-success barrier and improves the success rate from 0.00% to 9.00%. These results indicate that IFP is effective at eliciting the human-video-following capability learned from ICL data.

Effect of task-balanced robotic data. Finally, we validate that our task-level sampling strategy improves generalization in robotic pre-training, especially for cross-task transfer, by constructing a text-only Zero-WAM variant. In this variant, the ICL samples are still included during pre-training, but the human-video condition is masked, so each human-robot pair is reduced to an ordinary text-conditioned robotic video-action sample. This removes human video as an additional task specification signal and isolates the effect of task-balanced robotic pre-training data. We mainly compare with LingBot-VA, because most of our task-sampled robotic data are covered by the full pre-training corpus used by LingBot-VA. This makes LingBot-VA a natural reference for evaluating whether task-level re-partitioning and balanced sampling improve cross-task generalization beyond scaling the original pre-training distribution. Figure 6 (left) shows that the text-only Zero-WAM variant achieves a 39.44% seven-task average, outperforming LingBot-VA by 21.99 percentage points. The gains are especially clear on tasks such as *move stapler to pad*, *place bread in basket*, and *place empty cup*, indicating that task-balanced robotic pre-training improves cross-task transfer even without human-video conditioning.

5 Conclusions and Discussions

Conclusions. This work studies zero-shot cross-task generalization for robotic manipulation, where the policy must execute unseen tasks without collecting corresponding robot data or updating any model parameters. We approach this problem through causal video-action modeling and argue that the key to this setting is a

scalable in-context task interface: human video instructions generated at scale, supported by task-balanced data construction. To this end, we construct *HumanGen* through a scalable in-context human video generation pipeline that turns task-sampled robot trajectories into semantically matched human video instructions, and curate *Task-diverse* VA data from robotic pre-training data by sampling trajectories at the task level. Built on these data sources, Zero-WAM supports both language and human video demonstrations as task specifications within the same policy. The model is trained to predict future videos and executable actions, and is further shaped to follow in-context human videos for unseen task specification. Experiments in RoboTwin and real-world manipulation show that Zero-WAM can execute unseen tasks in simulation and unseen task configurations in the real world without any task-specific robot data or parameter updates, providing evidence that scalable human video instructions, combined with task-balanced robotic pre-training data, are a practical route toward generalizable robot manipulation.

Discussions. Zero-shot cross-task generalization is essential for deploying general-purpose robotic policies in open-ended real-world environments. Progress toward this goal requires both stronger transferable priors in robotic foundation policies and richer task interfaces that can convey intent across multiple modalities. Human demonstrations are a natural source of such task specifications, making continued growth in their scale and diversity important. Although our experiments focus primarily on stationary tabletop manipulation, future work should extend this paradigm to more complex, dynamic, and unstructured environments, including mobile manipulation and substantially longer-horizon tasks.

Among the available sources of transferable experience, egocentric human video is particularly promising because it can be collected at far greater scale than robot demonstrations. Its use, however, remains complicated by gaps in embodiment, observation, and action between humans and robots. The semantically aligned human-robot data introduced in this work may offer a bridge between abundant egocentric human video and comparatively scarce robot trajectories. By establishing correspondence at the level of task semantics rather than requiring exact motion-level alignment, this data formulation could allow robot policies to acquire broad task knowledge from human experience while relying on much less robot data for executable action supervision.

6 Related Works

6.1 Cross-Task Robotic Manipulation

Cross-task robotic manipulation evaluates whether a policy can execute an unseen manipulation task without collecting robot demonstrations for that task. Compared with visual generalization, such as changing scenes or object attributes around a seen task, cross-task generalization is more challenging because the model must infer previously unseen task-conditioned dynamics from the instruction and current observation.

Vision-language-action (VLA) models [42, 43, 3, 44, 5, 6, 45, 46, 47], such as $\pi_{0.5}$, provide a practical framework for scaling language-conditioned manipulation policies over heterogeneous robot data. However, VLA-style scaling has not closed the cross-task gap, partly because there remains a large mismatch between the vision-language representation space and the robot action space. AGNOSTOS [41] evaluates representative VLA policies under zero-shot cross-task manipulation, shows that they fail to effectively address this challenge, and mitigates it with a reference action-sequence strategy that uses a robot trajectory from the unseen task as in-context guidance. This workaround, however, requires providing an unseen-task robot trajectory at test time, thereby increasing the test-time cost.

Recently, world action models (WAMs) [48, 49, 50, 51, 52, 53, 54, 55, 56, 57] have offered a more promising direction: they shift the core challenge of unseen task generalization from direct action generalization to unseen video generation, since prior work [58, 59] has shown that accurate robot actions can be decoded from correctly predicted future videos.

6.2 World Action Models

World action models for robotics [11, 12, 13, 14, 60, 61, 15, 16, 62, 63, 64] jointly predict future visual states, such as videos or latent representations, and executable robot actions from an instruction and the current

observation. A major paradigm is video-action modeling. The LingBot-VA series [11, 12] develops this direction with autoregressive policies that couple causal video prediction with action prediction, and further studies native video-action pre-training for embodied control. DreamZero [13] evaluates WAMs as zero-shot policies under visual-domain shifts, showing that policy transfer can benefit from generated future dynamics rather than relying only on direct language-to-action prediction. EgoWAM [65] extends this view to egocentric human data, showing that world-model targets such as semantic features or 3D motion flow can improve human-to-robot transfer.

To make WAMs execute unseen tasks at test time, recent work explores test-time training. WAM-TTT [66] steers a frozen WAM by updating lightweight memory from raw human videos at deployment. RoboTTT [67] uses fast-weight test-time training to compress long execution histories into the policy state. These methods enable unseen-task execution at test time, but still require test-time adaptation through memory or fast-weight updates. In contrast, our Zero-WAM takes a different path: it builds human-robot ICL data and task-balanced video-action data into pre-training and post-training, so the policy can directly follow in-context human video instructions for zero-shot cross-task generalization at test time.

6.3 Human Video Data for Robotic Manipulation

Human videos have become an important data source for robotic manipulation because they contain diverse object interactions that are difficult to scale up with robots alone. Recent work uses large-scale in-the-wild human videos or human interaction demonstrations to learn robot-relevant visual representations, manipulation priors, or policy pre-training objectives, with varying degrees of grounding in robot actions [68, 69, 70, 71, 34, 72, 73, 74, 75]. Other methods move toward more task-aligned human data, including paired, retargeted, or jointly trained human and robot demonstrations and mixtures of in-the-wild and on-task human data [65, 76, 20, 77, 78, 79, 80], showing that task-aligned human-robot data can effectively reduce the human-robot domain gap. While these works mainly use human videos as training supervision or adaptation data, another set of methods directly conditions policies on human video demonstrations as task prompts [81, 17, 18, 82, 83, 84, 85, 86, 87, 88]. This makes human video a more direct task interface, and this line is closest to our use of in-context human video instructions. However, existing video-as-prompt systems either cover a limited set of tasks or require additional manual collection of human videos as task coverage grows. More broadly, across these uses of human video, zero-shot cross-task generalization still lacks a scalable data mechanism that jointly provides task diversity, human-robot semantic alignment, and executable robot action grounding: in-the-wild human videos provide broad behavior coverage but lack aligned robot data, task-aligned human-robot datasets are costly to scale across tasks, and policies conditioned on human video often depend on modular information extraction and alignment driven by manual priors.

Our Zero-WAM targets this missing data regime. Instead of collecting human demonstrations task by task, we automatically generate *HumanGen* data from task-sampled robotic video-action data. Because each generated human video is paired with a robot trajectory that retains executable actions, *HumanGen* scales human video instructions together with task-diverse robot dynamics, making human video task specification a scalable component of WAM training for zero-shot cross-task generalization.

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